Sujal Thapa

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SUMMARY

Mechanical engineering student interested in the intersection of real-world engineering problems and computational intelligence with a focus on **surrogate modeling** for conventionally expensive physics simulations. Experienced with gradient-enhanced Kriging, graph-neural-network surrogates, and CFD/FEM workflows in OpenFOAM and ANSYS.

PUBLICATION

Thapa, S., (first author) et al., ' Surrogate Modeling of Aeroelastic Response in a Hypersonic Intake'
Accepted, AIAA SciTech Forum 2027

EDUCATION

Institute of Engineering, Pulchowk Campus *Lalitpur, Nepal* 2022 – 2026
Bachelor of Engineering in Mechanical Engineering

- **Aggregate:** 78.5%
- **Relevant coursework:** Engineering Mathematics I–III, Finite Element Method, Statistics and Probability, Control Systems, Computer-Aided Design (CAD), Theory of Machines, Machine Design, Mechanical Vibration, Strength of Materials, Fluid Mechanics, Heat and Mass Transfer, Engineering Thermodynamics.

PROJECTS

Surrogate-Based Modeling for Aeroelastic Simulation of a Hypersonic Intake Apr 2025 – Mar 2026

- Developed a gradient-enhanced Gaussian process (Kriging) surrogate to model the total pressure recovery (TPR) response of a hypersonic intake across a range of deformation states.
  - Generated deformation states via Latin Hypercube Sampling and ran baseline CFD simulations for each.
  - Ran computationally efficient sensitivity simulations to acquire gradient information for the surrogate.
  - Constructed the gradient-enhanced Kriging (GEK) model and applied reduced-order modeling to reconstruct transient behavior from static solutions.
- **Skills:** CFD, sensitivity analysis, supervised learning, Gaussian process / gradient-enhanced Kriging surrogates.
- **Tools:** OpenFOAM, ANSYS ICEM CFD, Python (SMT, NumPy, pandas, matplotlib).

Stress Distribution Prediction of Cellular Beams using RegDGCNN Oct 2025

- Implemented a regression Dynamic Graph CNN (RegDGCNN) to predict stress distributions on cellular beams across varying hole diameters.
  - Prepared geometry and ran baseline FEM simulations of the cellular beams in ANSYS.
  - Extracted and cleaned simulation data in HDF5 (.h5) format for training.
  - Developed the RegDGCNN architecture and carried out validation and testing.
- **Skills:** FEM, graph neural networks, RegDGCNN, data pipeline construction.
- **Tools:** ANSYS, Python (NumPy, pandas, matplotlib).

Efficient Mesh Morphing Strategy for OpenFOAM Cases using IDW Jun – Sep 2025

- Wrote a Python tool using inverse distance weighting to morph meshes automatically, avoiding manual geometry perturbation and re-meshing.
  - Extracted geometry and patch points from the OpenFOAM case structure in Python.
  - Applied modeled deformation to target patches, then propagated it to interior mesh points via inverse distance weighting.
  - Updated the mesh files to reflect the morphed configuration.
- **Skills:** mesh manipulation, file handling in Python, OpenFOAM case structure.
- **Tools:** OpenFOAM, Python.

Gaussian Process Regression for Hypersonic Wedge Flow Jun – Aug 2025

- Benchmarked the wall-clock savings of a gradient-enhanced surrogate built on sensitivity-simulation gradients over a baseline-only surrogate.
  - Designed the experiment via Latin Hypercube Sampling and selected step sizes for the sensitivity simulations.
  - Performed exploratory data analysis and compared ordinary Kriging (OK) and gradient-enhanced Kriging (GEK) models.
- **Skills:** hypersonic CFD, ordinary and gradient-enhanced Kriging, warm-start optimization.
- **Tools:** OpenFOAM, ANSYS, Python (SMT, NumPy, pandas, matplotlib).

EXPERIENCE

Accelerated Komputing *Pittsburgh, PA (Remote)* Aug 2026– Present

Research Engineer

- Worked on Neural-operators, graph neural networks (GNNs), and reduced-order models.

Accelerated Komputing *Pittsburgh, PA (Remote)* Oct 2025 – March 2025

Research Intern / On-the-Job Trainee

- Developed a RegDGCNN surrogate model for structural stress prediction on cellular beams (see Projects)

MeroSpace, SOMAES — Pulchowk Campus *Lalitpur* 2023 – 2025

Editor

- Wrote and edited technical articles and publications for the student engineering society.

Pulchowkians' Fitness Hub *Lalitpur* 2023 – 2024

Marketing Lead / Secretary

- Led marketing and outreach; coordinated a small team.
- Managed logistics for college-wide events.

TECHNICAL SKILLS

- **Statistical / surrogate modeling:** regression models, ordinary Kriging, gradient-enhanced Kriging, reduced-order modeling, graph neural networks.
- **Programming:** Python (proficient), MATLAB, C.
- **Computational tools:** OpenFOAM, ANSYS, ParaView, FreeCAD.
- **Other:** technical writing and editing, collaboration, communication.

SCHOLARSHIP

- **Government Merit Scholarship, IOE Entrance Examination**
  - Awarded a fully funded government seat at Pulchowk Campus by ranking 232 among 15,000–20,000 candidates nationwide (top 1.5%) in Nepal's national engineering entrance examination.